



Age-Related Macular Degeneration

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Introduction

Age-related macular degeneration (AMD) is classified as dry (nonexudative) or neovascular (wet or exudative). Dry AMD is characterized by drusen (white to yellow spots in the central retina) and may or may not be associated with gradual deterioration in central vision. Neovascular AMD (nAMD) always occurs in the setting of pre-existing dry AMD and is responsible for the vast majority of severe vision loss. nAMD is caused by the presence of a choroidal neovascular membrane (CNV), a collection of abnormal blood vessels that grows underneath, and sometimes into, the retina. It leaks fluid into and under the retina, may bleed into the nearby tissue, and eventually creates a scar, resulting in permanent vision loss. This vision loss is generally limited to the central eyesight, which is critical for reading, driving, recognizing faces and performing fine visual tasks. It rarely affects the peripheral or side vision and does not usually cause total blindness. Patients who have severe nAMD affecting both eyes can generally walk around unaided but may require assistance for activities of daily life requiring visual precision, e.g., navigating traffic lights, seeing bus numbers. These patients may qualify as legally blind due to poor vision. There are a number of similar conditions that cause central visual loss, the most common of which is pathologic myopia.

In the Western world, almost one-third of people over 75 years of age develop dry AMD^[1] and 5% of these individuals will go on to develop nAMD. About 90% of severe vision loss occurs in those patients with nAMD.

Up to 30% of the risk of vision loss in AMD is attributable to smoking.^[2] Hypertension,^{[2][3][4]} family history of AMD,^[2] and light pigmentation of the skin and iris are also risk factors. Vision loss from nAMD in one eye is associated with a 50% risk of vision loss in the fellow eye over 5 years.^{[5][6][7]} Nutrition (aside from eating a balanced diet) and wearing sunglasses do not appear to have a significant impact on visual outcomes in AMD.

Goals of Therapy

- Minimize loss of vision
- Improve vision if possible
- Minimize enlargement of central scotomas (central blur in centre of visual field)
- Optimize function and quality of life

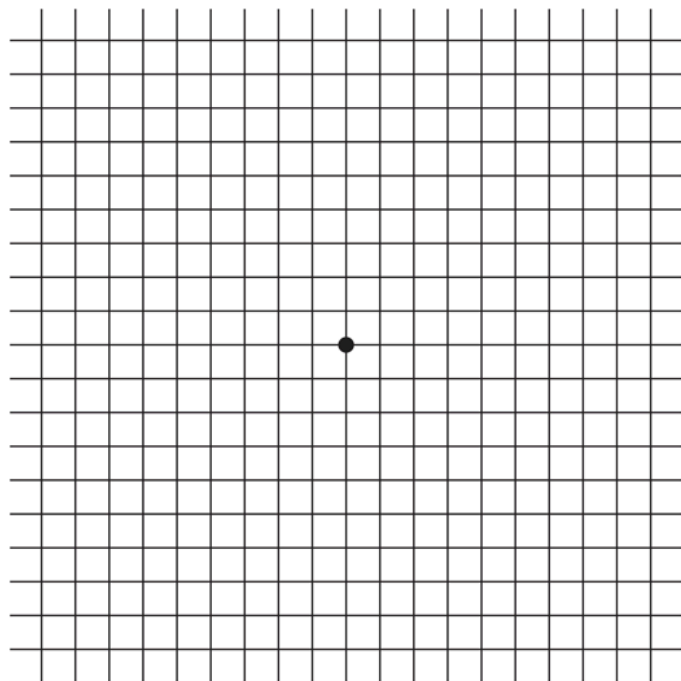
Investigations

- Thorough history with attention to:
 - vision loss
 - recent changes
 - central visual changes such as a central scotoma
 - metamorphopsia (distortion such that straight lines appear curved or crooked)
 - AMD in patient or first-degree relatives
- Physical examination with attention to:
 - visual acuity
 - Amsler grid (see [Figure 1](#))
 - funduscopy (dilated fundus exam with slit-lamp biomicroscopy)

Note: The retinal exam alone may be inadequate to exclude the presence of a treatable lesion in nAMD.
- Optical coherence tomography (OCT):
 - widely available
 - noninvasive
 - uses infrared light to generate a cross-sectional view of the macula
 - considered the gold standard for treatment decisions using vascular endothelial growth factor inhibitors
- Optical coherence tomography angiography (OCT-A):
 - uses the same machine as OCT but offers additional information
 - used as an adjunct to conventional OCT but does not replace it
- Fluorescein angiography:
 - available only at large centres and interpreted by retinal specialists
 - involves IV injection of fluorescein sodium, followed by photography of the ocular fundus
 - considered the gold standard for investigating many retinal diseases

Note: Some patients have allergic reactions to fluorescein sodium, which is chemically unrelated to x-ray contrast and contains no iodine. Severe reactions are rare.

Figure 1: Amsler Grid



1. The size of the grid is generally standardized at 10 cm x 10 cm and it should be held 33 cm away from the patients's eyes. This is important as the size and the distance determine the corresponding area in the retina that is being tested. Suggest patients post the grid in a prominent location (e.g., on the refrigerator) to remind them to test weekly.

1. Instructions for the patient:

- Use adequate lighting and wear the glasses you would normally use to read.
- Stand 33 cm away from the grid.
- Cover one eye at a time and test the other.
- With the open eye, stare at the centre dot while considering the following questions:
 - Is any corner or border of the grid not visible?
 - Are there any broken or missing lines?
 - Are any of the lines blurry, curved or wavy?
- Repeat the test once a week. Tell your eye-care professional if you answer yes to any of the questions.

1. Available from: www.allaboutvision.com/conditions/amsler-grid.pdf.

Therapeutic Choices

An algorithm for the management of age-related macular degeneration is presented in [Figure 2](#).

Nonpharmacologic Choices

Smoking is a major risk factor for AMD. Advise patients to quit smoking (see [Tobacco Use Disorder: Smoking and Nicotine Cessation](#)) and to maintain a balanced diet. See also [Prevention of Progression](#).

The Amsler grid (see [Figure 1](#)) is useful for self-monitoring by patients with either nAMD or dry AMD and/or a family history of AMD.^[8] A central blur or distortion in either eye can indicate the onset of nAMD. The test must be performed with one eye closed, then repeated with the other eye. Reading glasses must be used for the test if the patient customarily requires them. The grid should be placed 33 cm from the eyes.

Visual aids such as magnifiers, high-power glasses and reading machines can enable many visually handicapped persons to read enough to perform common household tasks. Low-vision clinics sponsored by the Canadian National Institute for the Blind are widely available in Canada.

Pharmacologic Choices

Medications used in the management of patients with AMD are listed in [Table 2](#).

Vascular Endothelial Growth Factor Inhibitors

Studies have proven that the vascular endothelial growth factor (VEGF) inhibitors are effective in decreasing vision loss in nAMD.^{[9][10][11][12]} By decreasing VEGF levels in the extracellular space, these drugs inhibit growth and cause regression of new blood vessels in CNV membranes. They also stabilize the blood-retinal barrier, thereby reducing macular edema and restoring normal anatomy. All VEGF inhibitors are injected directly into the vitreous cavity of the eye following application of topical anesthesia and antiseptic eye drops, usually povidone-iodine. Physicians should inquire about allergy to iodine prior to instillation. Alternatives such as chlorhexidine topical solution are currently available in Canada.

The VEGF inhibitors available in Canada (and their entry to market): bevacizumab (2005), ranibizumab (2007), aflibercept (2013), brolucizumab (2020) and faricimab (2022).

Bevacizumab is a monoclonal antibody that has been in widespread use since 2005 when it was approved for the treatment of colon cancer. It is frequently used off-label for several retinal diseases, including nAMD.

Ranibizumab is a monoclonal antibody fragment derived from bevacizumab that demonstrates a comparatively higher VEGF binding affinity.^{[11][12]} Ninety percent of patients receiving ranibizumab lost fewer than 3 lines on a specialized vision chart and 40% of patients showed improvement at 2 years.^[11] Large-scale, multicentre, controlled clinical trials have shown no statistically significant differences between ranibizumab and bevacizumab for visual outcomes of patients with nAMD;^{[13][14][15][16][17]} however, minor differences in other outcome measures, such as intraretinal edema and atrophy, were reported. There are small differences in the half-lives of the 2 drugs and tachyphylaxis has been reported with both, with improvements upon switching to the other drug. Given the very large difference in cost and minimal evidence to show that one is superior to the other, most physicians will initiate treatment with bevacizumab.

Aflibercept is a fusion protein of binding domains from human VEGF receptors 1 and 2 with the constant region (Fc) of human immunoglobulin G. Two large-scale, multicentre, controlled clinical trials showed equivalent efficacy at 96 weeks for aflibercept and ranibizumab in the treatment of nAMD.^[18] Aflibercept is currently in widespread use as a second-line therapy, generally following an insufficient response to bevacizumab. It appears to offer certain patients decreased intraretinal and subretinal fluid, decreased height of pigment epithelial detachment, and increased treatment intervals.

Brolucizumab binds to VEGF via a single-chain antibody fragment. Large multicentre, controlled clinical trials have demonstrated non-inferiority to aflibercept.^[19] Unfortunately, brolucizumab appears to have a much higher rate of intraocular inflammation at 4.6% versus 1.1% for aflibercept, and an associated higher rate of vision loss at 0.74% versus 0.14% for aflibercept.^[20] A 2020 report of multiple cases of retinal vasculitis with poor visual outcomes suggests that the inflammatory side effects of this new drug extend to the retina itself, rather than being limited to uveitis alone.^[21] Concerns over these adverse effects appear to be limiting the clinical use of brolucizumab.

Faricimab is a bispecific antibody that acts through dual inhibition of both angiopoietin-2 and vascular endothelial growth factor A (VEGF A). Combined results of two large, multicentre trials demonstrated non-inferiority to aflibercept and a comparable profile of adverse events. Overall ocular and systemic serious adverse events were $\leq 2\%$ of the study population. The 48-week results showed 80% of patients could be treated on an extended schedule of 3–4 months, including nearly 45% of patients treated every 4 months.^[22] Since the majority of patients treated with previously-established medications for nAMD require monthly dosing, faricimab has the potential to ease the burden of treatment for both patients and health-care professionals. To date, there is little clinical experience with this drug outside the research trials and its precise role in therapy remains to be determined.

Despite small differences in the effects of various anti-VEGF drugs in large trials, individual patients may respond differently. In clinical practice, if the response to one agent is felt to be inadequate (as judged by lack of improvement in markers of pathology on OCT scans), patients may be switched to another anti-VEGF agent. The mechanism of action of all the drugs is similar, and a particular patient's response cannot be predicted. Indeed, patients are sometimes switched back to the original agent with satisfactory clinical outcomes. At least 25% of patients appear to require switching from one anti-VEGF drug to another at some point in their course of treatment.^[23]

The frequency of administration of bevacizumab, ranibizumab and aflibercept is most commonly every 4 weeks. It may be possible to extend the time between treatments further with brolucizumab, which appears to be its chief advantage over other drugs in its class, and faricimab has proven effective at intervals of 12–16 weeks. While the endpoints of treatment with AMD have not been fully established, most patients will require a year or more of therapy. Some can be put on a “treat-and-extend” protocol earlier, while others will require ongoing treatment as long as there is evidence of active disease on OCT. Patients who fail cessation of treatment multiple times may be at risk of vision loss if treatment is discontinued.

Side effects of these drugs are uncommon. The most worrisome is due to the injection procedure—bacterial endophthalmitis occurs at a frequency of 3 in 10 000 procedures or less.^[24] Severe loss of vision is the hallmark of endophthalmitis. Pain and redness of the eye may also be present. Because irreversible blindness may occur, endophthalmitis is considered a medical emergency that requires immediate treatment by an ophthalmologist; management includes biopsy for bacterial culture followed by intravitreal injection of antibiotics. Often surgery is necessary, and outcomes vary widely depending on the duration and nature of the infection. Patients receiving intravitreal VEGF inhibitors are warned to seek medical attention immediately if they develop symptoms of endophthalmitis.

Many patients will have mild pain from the drug-injection procedure, which does not necessarily indicate a problem. Subconjunctival hemorrhage is also common and generally resolves within 2 weeks. A large fraction of patients undergoing intraocular injection will have a subconjunctival hemorrhage, usually manifested as a small red spot at the injection site on the white of the eye. Generally the spot will go away within a few days and has no effect on the inside of the eye or on any macular disease. Patients may shower or bathe normally afterward. Subconjunctival bleeding is more common in patients taking ASA or anticoagulants, and patients with a propensity for bleeding should discuss this issue with their doctor prior to treatment.

The recurrence rate of AMD following successful treatment has not been firmly established but appears to be high. Lifelong vigilance for recurrence is likely necessary, including self-monitoring with the Amsler grid (see [Figure 1](#)).

Anti-VEGF therapy has also become the standard treatment of diabetic macular edema, branch and central retinal vein occlusion with macular edema, and various other retinal diseases.

Photodynamic Therapy

Photodynamic therapy (PDT) involves IV infusion of **verteporfin** followed by administration of laser light to the affected eye. Verteporfin is a porphyrin derivative that is activated by laser light to generate free radicals that damage the neovascular endothelium and occlude the blood vessels. In the treatment of AMD, PDT has been shown to significantly reduce vision loss compared with placebo;^{[26][27][28]} however, patients usually do not recover lost vision and often lose further vision after the initiation of therapy before plateauing at levels higher than untreated patients. Verteporfin causes photosensitivity that can result in severe burns. Patients must avoid exposure of skin or eyes to sunlight or bright indoor light for 2 days following treatment.

PDT is no longer used as primary therapy for most forms of AMD as it is much less effective and more complicated to administer compared with the VEGF inhibitors. PDT may be indicated in patients with a suitable CNV lesion from rare forms of nAMD or related conditions.^{[25][26][27][28]} Currently, PDT is most commonly used for the treatment of central serous retinopathy, also known as central serous chorioretinopathy. This condition usually presents in young males, is not caused by macular degeneration and has a more benign course. PDT may also be used to treat polypoidal choroidal vasculopathy (PCV), a condition that shares many similarities with AMD and can be diagnosed using a procedure similar to fluorescein angiography called indocyanine green angiography (ICGA).

Prevention of Progression

Patients should be advised to quit smoking to reduce their risk of AMD (see [Tobacco Use Disorder: Smoking and Nicotine Cessation](#)). Following a balanced diet should be encouraged.

Numerous vitamin and mineral supplements are marketed to help reduce the risk of AMD. Choosing a product with safe ingredients in effective strengths requires careful assessment of the evidence (see [Table 1](#)).

Table 1: Vitamin and Mineral Supplements for Prevention of AMD Progression

Supplement Daily Dosage	AREDS ^[29]	AREDS2 ^[32]	Comment
beta-carotene 25 000 units	25% reduction of AMD-associated vision loss over 7 years (AREDS combination)	Formulation without beta-carotene as effective as formulation containing beta-carotene	Not recommended for prevention of progression to AMD. ^[32] Beta-carotene was associated with an increased risk of lung cancer regardless of current or past smoking status. ^{[30][31]} Contraindicated in severe liver disease, renal insufficiency, renal dialysis and pregnancy.
vitamin C 500 mg	25% reduction of AMD-associated vision loss over 7 years (AREDS combination)	See AREDS	High-dose supplementation contraindicated in renal disease; may cause oxalosis resulting in kidney stones.

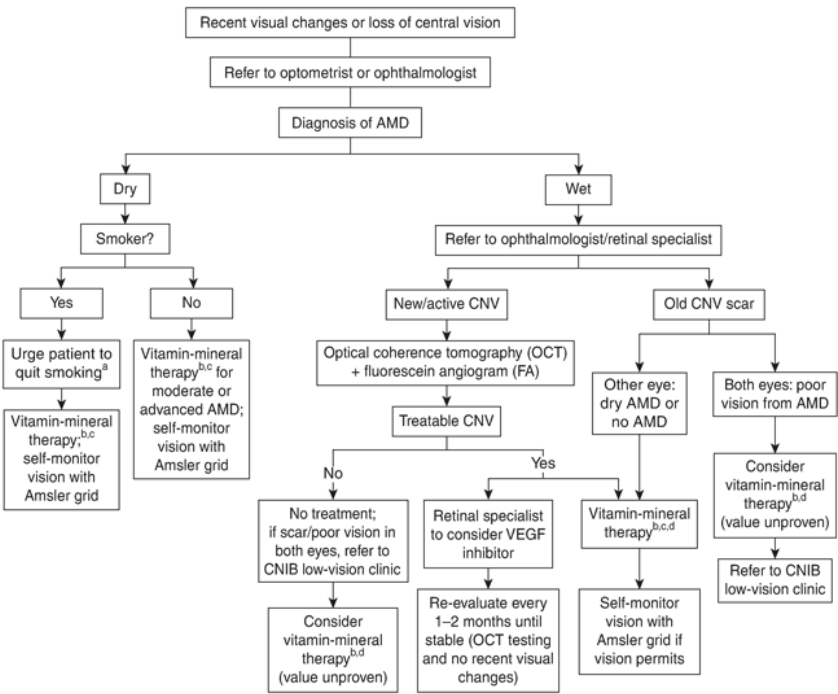
Supplement Daily Dosage	AREDS ^[29]	AREDS2 ^[32]	Comment
vitamin E 400 units	25% reduction of AMD-associated vision loss over 7 years (AREDS combination)	See AREDS	High-dose supplementation contraindicated in renal disease. May exacerbate retinitis pigmentosa; contraindicated in certain non-AMD inherited retinal degenerations. ^[33]
copper 2 mg	25% reduction of AMD-associated vision loss over 7 years (AREDS combination)	See AREDS	Included to reduce the likelihood of anemia in patients taking large amounts of zinc.
zinc 80 mg	High-dose formulation (80 mg): 25% reduction of AMD-associated vision loss over 7 years (AREDS combination)	N/A	High-dose supplementation may cause gastric disturbances and the rare, severe, sideroblastic form of anemia.
zinc 25 mg	N/A	Low-dose formulation (25 mg): as effective as high-dose formulation and may cause less nausea	
lutein 10 mg + zeaxanthin 2 mg	N/A	Addition did not further reduce the risk of progression to advanced AMD	Structurally and functionally related to beta-carotene and vitamin A, occur naturally in food and are concentrated in the macula; may be an effective and safer alternative than beta-carotene, although more research is required.

Supplement Daily Dosage	AREDS ^[29]	AREDS2 ^[32]	Comment
omega-3 fatty acids (DHA 350 mg + EPA 650 mg)	N/A	Addition did not further reduce the risk of progression to advanced AMD	

Abbreviations: AMD = age-related macular degeneration; AREDS = age-related eye disease study; AREDS2 = age-related eye disease study 2; DHA = docosahexaenoic acid; EPA = eicosapentaenoic acid; N/A = not applicable

Algorithms

Figure 2: Management of Age-Related Macular Degeneration



[a] See [Tobacco Use Disorder: Smoking and Nicotine Cessation](#).
[b] Vitamin-mineral formulations containing beta-carotene are no longer recommended, see [Table 3](#).
[c] See [Pharmacologic Choices](#).
[d] If patient smokes, encourage smoking cessation, see [Tobacco Use Disorder: Smoking and Nicotine Cessation](#).

Abbreviations: AMD = age-related macular degeneration; CNIB = Canadian National Institute for the Blind; CNV = choroidal neovascular membrane; OCT = optical coherence tomography; VEGF = vascular endothelial growth factor

Drug Tables

Table 2: Drug Therapy for Neovascular Age-Related Macular Degeneration

Drug/Cost ^[a]	Dosage	Comments
Drug Class: Photodynamic Therapy		

Drug/Cost ^[a]	Dosage	Comments
verteporfin Visudyne ~ \$1,900/15 mg vial	6 mg/m ² diluted to a total volume of 30 mL and given IV over 10 min; may need to be repeated at 3-month intervals	Use only in rare forms of nAMD. IV infusion is followed by laser light delivery, i.e., 50 J/cm² at a fluence of 600 mW/cm² (standard fluence); 25 J/cm² at a fluence of 300 mW/cm² (low fluence) is also used. Patients must avoid exposure of skin or eyes to sunlight or bright indoor light for 2 days. Severe back pain, which occurs rarely during verteporfin infusions, remits when the infusion is stopped. Severe and potentially irreversible vision loss occurs rarely following therapy.

Drug Class: Vascular Endothelial Growth Factor Inhibitors

afibercept Eylea ~\$1,500/2 mg vial	2 mg intravitreal injection every 4 wk initially; after improvement or stabilization on OCT , a treat-and-extend protocol may be used	VEGF Inhibitors: mild eye pain is common; does not necessarily indicate a problem. Subconjunctival hemorrhage is also common and resolves spontaneously. Bacterial endophthalmitis occurs rarely but is a medical emergency requiring immediate treatment by an ophthalmologist.
bevacizumab	1.25 mg intravitreal injection every 4 wk	VEGF Inhibitors: mild eye pain is common; does not

Drug/Cost ^[a]	Dosage	Comments
Avastin, Abevmy ^[b] Aybintio ^[b] Bambevi ^[b] Mvasi ^[b] Zirabev ^[b] Avastin: \$550/100 mg vial Biosimilars: \$370/100 mg vial	initially; after improvement or stabilization on OCT , a treat-and-extend protocol may be used	necessarily indicate a problem. Subconjunctival hemorrhage is also common and resolves spontaneously. Bacterial endophthalmitis occurs rarely but is a medical emergency requiring immediate treatment by an ophthalmologist. Bevacizumab: widely used off-label.

brlucizumab Beovu ~\$1,450/6 mg vial	6 mg intravitreal injection every 4 wk for the first 3 doses; treatment interval every 8 or 12 wk thereafter based upon disease activity	VEGF Inhibitors: mild eye pain is common; does not necessarily indicate a problem. Subconjunctival hemorrhage is also common and resolves spontaneously. Bacterial endophthalmitis occurs rarely but is a medical emergency requiring immediate treatment by an ophthalmologist.
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faricimab Vabysmo ~\$1,425/6 mg vial	6 mg intravitreal injection every 4 wk for the first 4 doses; treatment interval of 8, 12 or 16 wk thereafter based upon anatomic and visual acuity evaluations at Week 20 and Week 24	VEGF Inhibitors: mild eye pain is common; does not necessarily indicate a problem. Subconjunctival hemorrhage is also common and resolves spontaneously. Bacterial endophthalmitis occurs rarely but is a medical emergency requiring immediate
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Drug/Cost ^[a]	Dosage	Comments
		treatment by an ophthalmologist.
ranibizumab Lucentis, Byooviz ^[b] ~\$1,750/2.3 mg vial Biosimilars: ~\$1,050/100 mg vial	0.5 mg intravitreal injection every 4 wk initially; after improvement or stabilization on OCT , a treat-and-extend protocol may be used	VEGF Inhibitors: mild eye pain is common; does not necessarily indicate a problem. Subconjunctival hemorrhage is also common and resolves spontaneously. Bacterial endophthalmitis occurs rarely but is a medical emergency requiring immediate treatment by an ophthalmologist.

^[a] Includes drug cost only.

^[b] Biosimilar.

Abbreviations: AMD = age-related macular degeneration; OCT = optical coherence tomography; VEGF = vascular endothelial growth factor

Table 3: Vitamin–Mineral Supplements for Age-Related Macular Degeneration

Drug ^[a] /Cost ^[b]	Dosage	Comments
Drug Class: Combination Vitamin/Mineral Supplements		
vitamin C 250 mg/vitamin E 200 units/copper 1 mg/zinc 40 mg/lutein 5 mg Vitalux Advanced, Vitalux Advanced Chewable, Vitalux-S, generics \$10–15	1 caplet BID PO	Formulated to match the doses in the AREDS2 trial. ^[32] Take with food to reduce nausea that may be associated with zinc dose. Note: contains no beta-carotene. Formulas containing beta-carotene may increase the risk of lung cancer. Lutein 5 mg content yields zeaxanthin 250 mcg.

Drug ^[a] /Cost ^[b]	Dosage	Comments
		Caution with other vitamin–mineral supplements containing overlapping ingredients.

^[a] Because of an increased risk of lung cancer, mainly in people who smoke^{[30][31]} but also in people who formerly smoked or who don't smoke,^[32] formulations containing beta-carotene are not recommended.

^[b] Cost of 30-day supply; includes drug cost only.

Suggested Readings

[Age-Related Eye Disease Study 2 Research Group. Lutein + zeaxanthin and omega-3 fatty acids for age-related macular degeneration: the Age-Related Eye Disease Study 2 \(AREDS2\) randomized clinical trial. JAMA 2013;309\(19\):2005-15.](#)

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